Software Requirements Specification

Vehicle Parking Management and Tracking System

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1. Functional Requirements

ID	Requirement	Priority
IR-FR-01	The system shall allow users to register an account and log in using valid credentials.	High
IR-FR-02	The system shall allow users to register, view, update, and manage their vehicle information.	High
IR-FR-03	The system shall allow users and authorized staff to view current parking-slot availability and status.	High
IR-FR-04	The system shall allow authenticated users to reserve an available parking slot for a specified period.	High
IR-FR-05	The system shall allow authorized staff to record vehicle entry and create an active parking session.	High
IR-FR-06	The system shall assign an available parking slot to a vehicle and prevent duplicate slot assignments.	High
IR-FR-07	The system shall allow authorized users to identify the current location and status of a parked vehicle.	High
IR-FR-08	The system shall allow authorized staff to record vehicle exit and complete the parking session.	High
IR-FR-09	The system shall calculate parking duration and fees according to configured parking rates.	High
IR-FR-10	The system shall allow users or staff to make and record parking payments.	High
IR-FR-11	The system shall maintain and display historical parking-session information.	Medium
IR-FR-12	The system shall allow administrators to manage users, vehicles, slots, rates, and generate reports.	High

2. Non-Functional Requirements

ID	Category	Requirement
IR-NFR-01	Performance	Display parking-slot availability within 2 seconds for at least 95% of requests.
IR-NFR-02	Response Time	Process vehicle entry/exit transactions within 3 seconds, excluding payment processing.

ID	Category	Requirement
IR-NFR-03	Availability	Maintain at least 99% availability during defined operating hours.
IR-NFR-04	Scalability	Support at least 500 slots, 5,000 registered vehicles, and 100 concurrent users.
IR-NFR-05	Usability	An attendant shall complete a normal entry transaction within 1 minute after basic training.
IR-NFR-06	Reliability	Maintain consistent slot and session data under simultaneous parking operations.
IR-NFR-07	Maintainability	Built using modular components for users, vehicles, parking, payment, and reporting.
IR-NFR-08	Portability	Support common web browsers such as Chrome, Edge, and Firefox.
IR-NFR-09	Data Integrity	Maintain accurate relationships between users, vehicles, slots, sessions, and payments.

3. Security Requirements

ID	Requirement
IR-SEC-01	Authentication required before accessing protected functions.
IR-SEC-02	Role-based authorization for Drivers, Attendants, and Administrators.
IR-SEC-03	Passwords stored using a secure hashing mechanism.
IR-SEC-04	Account temporarily locked after 5 consecutive failed logins.
IR-SEC-05	Inactive authenticated sessions terminated after 15 minutes.
IR-SEC-06	All user input validated before processing or storing.
IR-SEC-07	Audit logs maintained for key administrative activities.
IR-SEC-08	Only authorized administrators may access audit logs.
IR-SEC-09	Unauthorized users prevented from accessing confidential user/payment data.